Open Ecosystem Can Unlock Velocity for AI DCs

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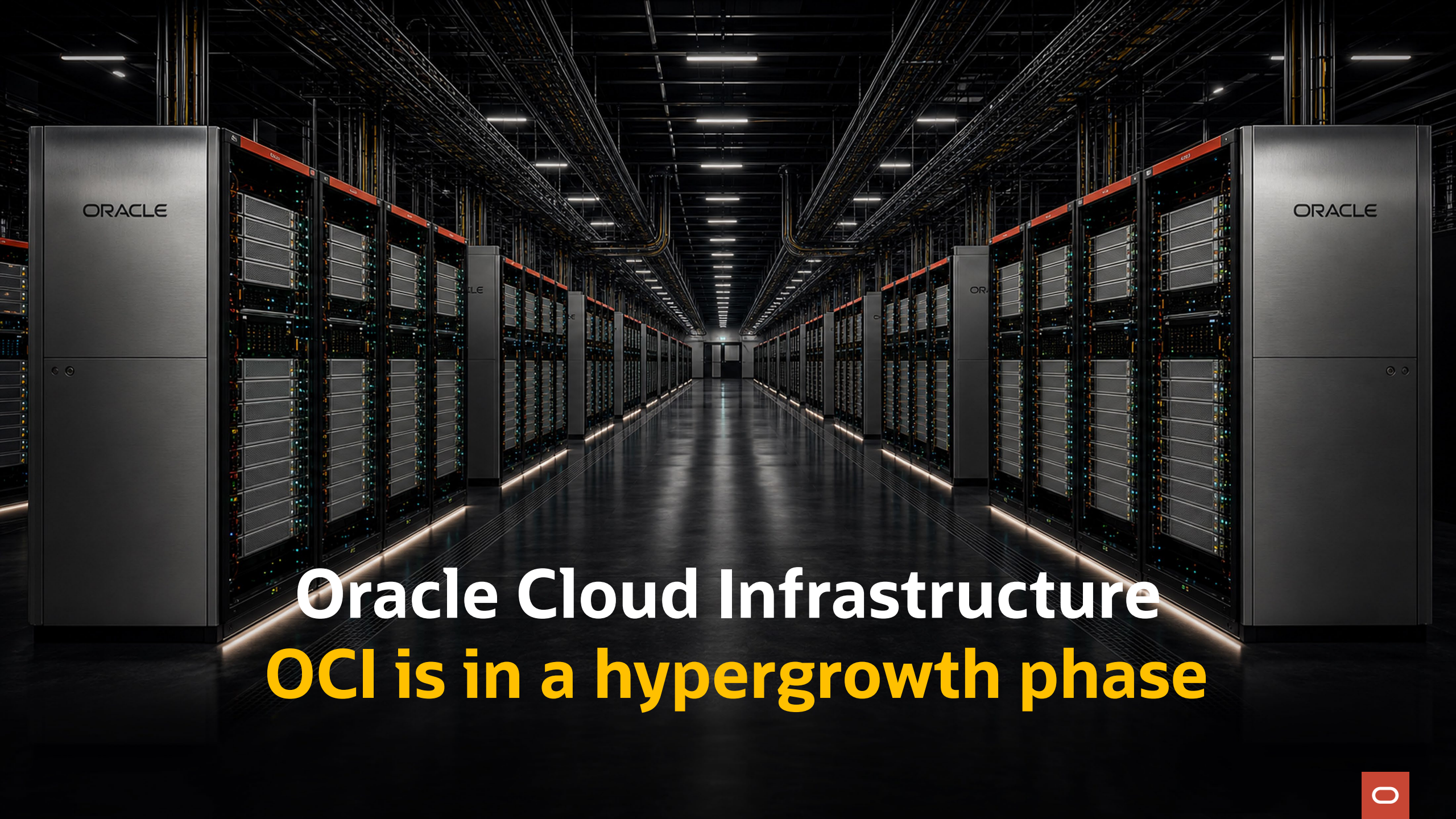
OCP GLOBAL SUMMIT

October 12-15, 2026
San Jose, California



Oracle Cloud Infrastructure

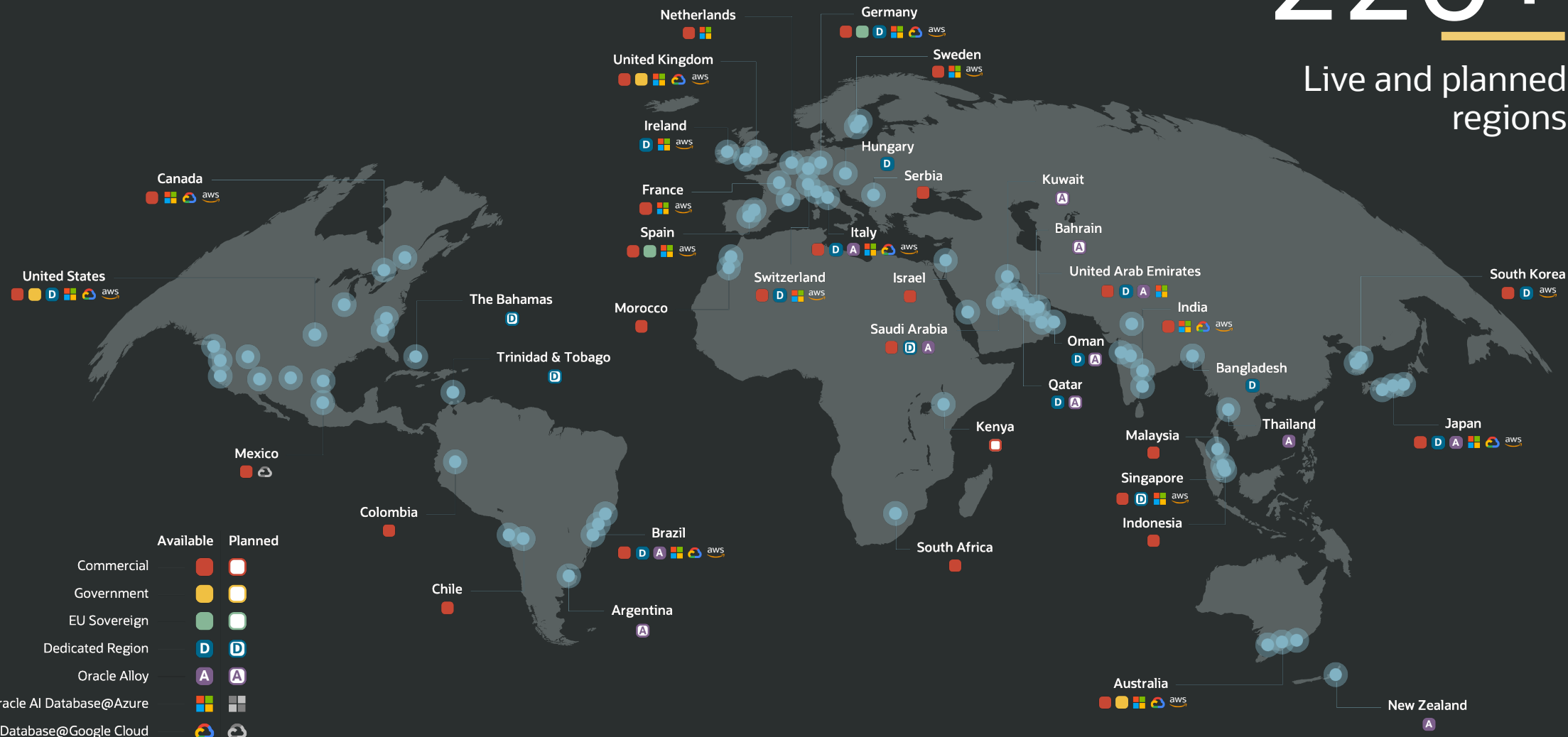
OCI is in a hypergrowth phase

Oracle Cloud Infrastructure

Different regions for different customers

220+

Live and planned regions



Expanding Responsibly

OCI is committed to responsibly building and operating AI data centers for the communities we serve.

COMMITMENT TO COMMUNITY

Prosperity for Data Center Communities

- Investing in the local community
- Reducing Water Consumption
- Independent Microgrid
- Workforce Development

ORACLE + OPEN Compute Project®

Empowering communities through an Open Ecosystem

- Delivering interoperability at data-center scale
- Enabling responsible growth
- Turn operating experiences into industry progress

OCI Leadership across OCP & Open ecosystem

2021
OCI Engaged in

- DC-SCM v2

2023
AI Infrastructure Surge

- Started adoption of OCP OR v3

2025
Several Gigawatt DC sites for next-gen AI

- OR v3 OCI C3 racks
- Engagement across multiple workgroups

2026
Responsible AI Infrastructure

- Initiating several new OCP Workgroups
- Key Leadership Contributions with Specifications
- OCP Advisory Board Member

OCI Contributions to OCP Initiatives

OCI Leadership

Sustainability, Efficiency & Communities

NEW

Tessera - Integrated Hardware Security Module

Optics Reliability & Stability

NEW

DPU Abstraction Layer

Sustainability & Efficiency

DC Efficiency, Sustainability metrics & Communities

Responsible Sustainability Requires Contextual Decisions

NET-ZERO IMPACT

Efficient
Power & Cooling

Holistic Sustainability
Metrics

Local Context
Matters

Best-Practice
Guidance

NEWLY PROPOSED

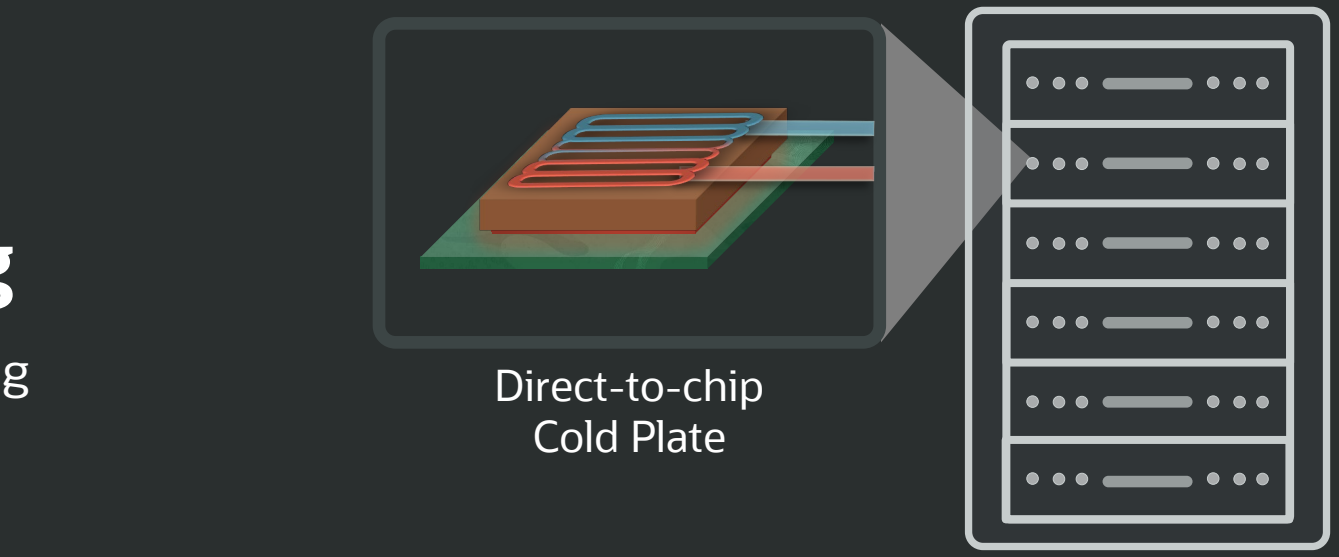
Data Center Efficiency Metrics

Whitepaper - Geographic PUE (gPUE) metrics, Compute Density metrics

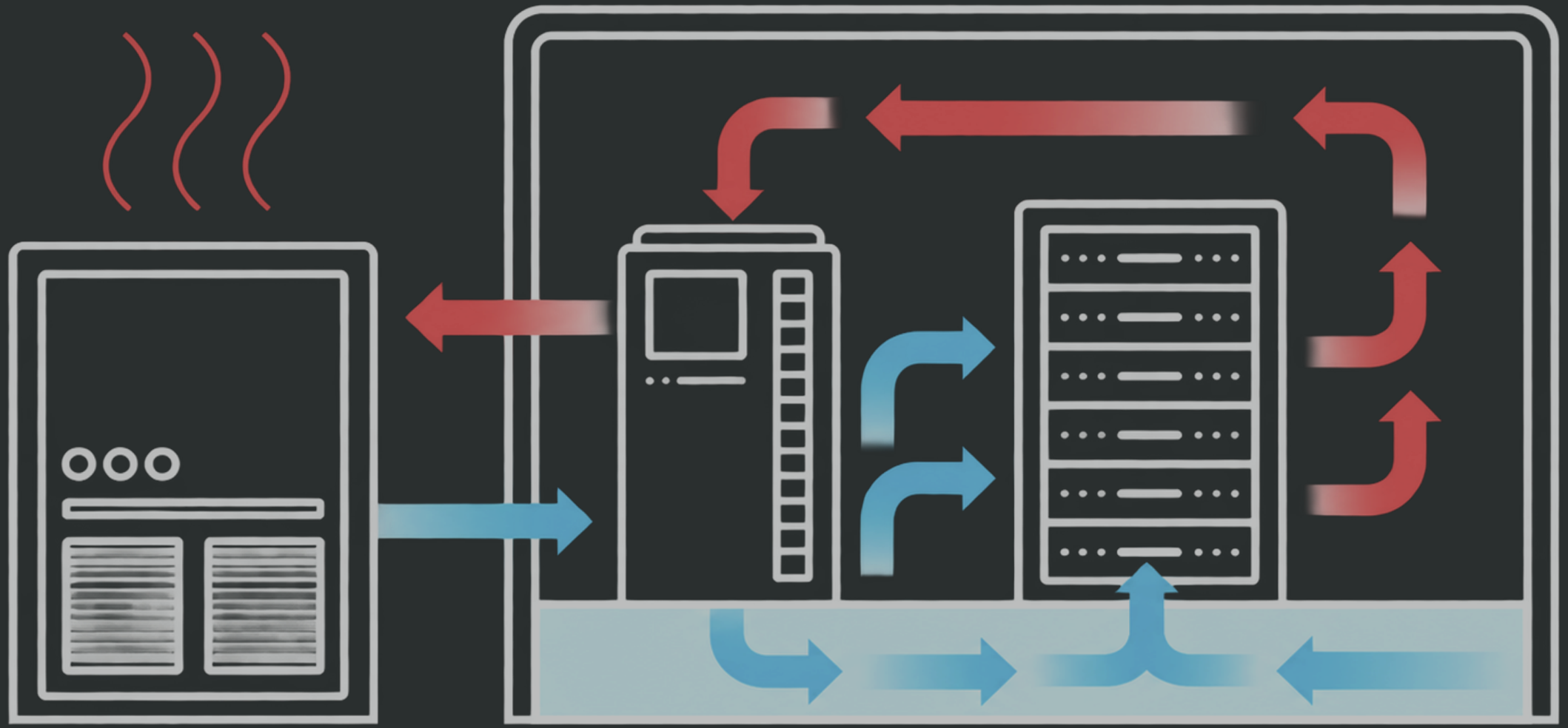
PUE and Infrastructure Utilization Efficiency (IUE) Metrics of Energy Efficiency from Servers to Data Centers

Focus on Next-gen Cooling

AI heat density keeps accelerating



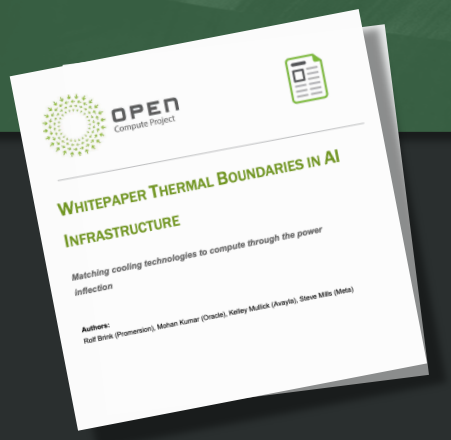
Whole-system plumbing compounds complexity

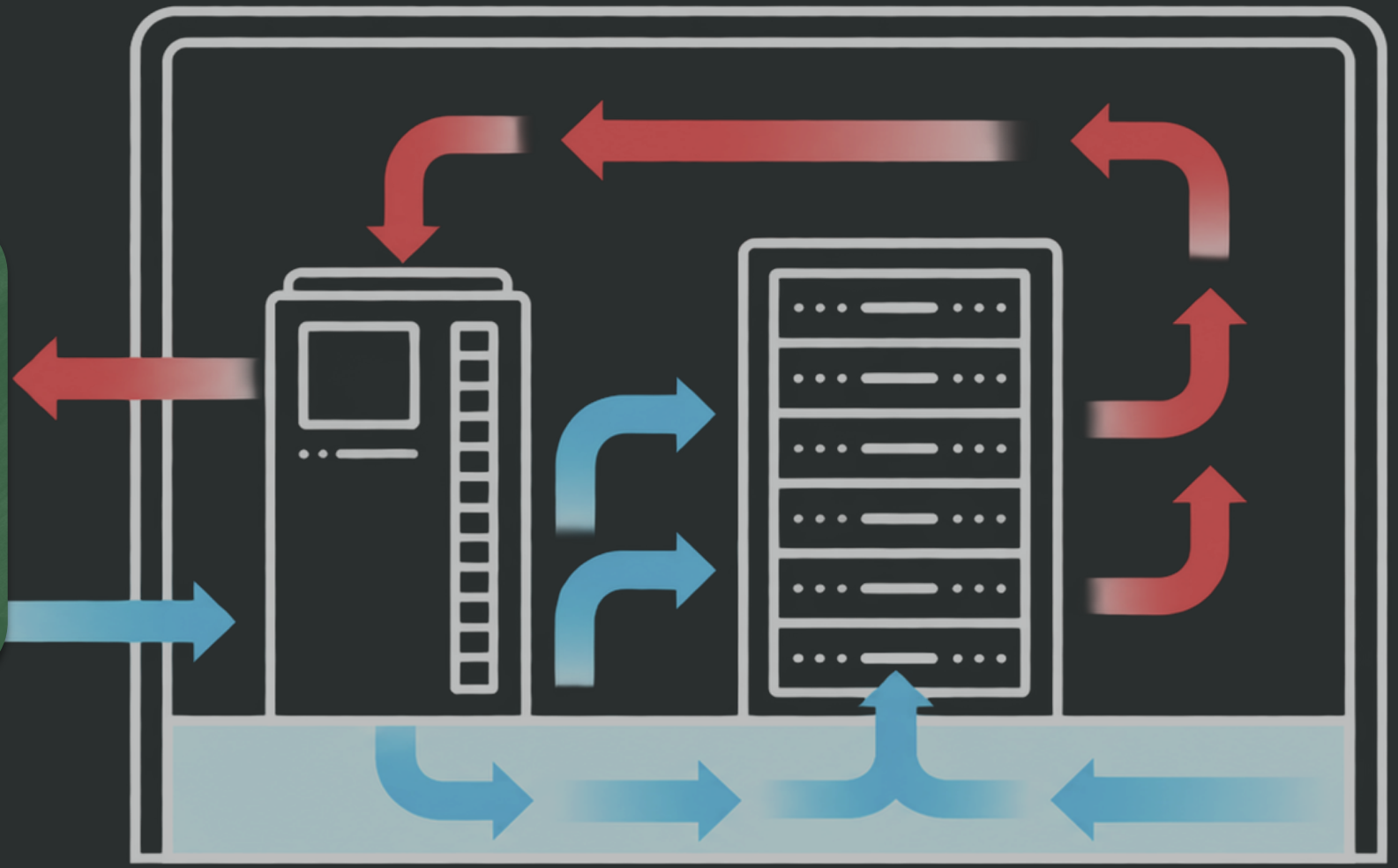


CALL TO ACTION

JOIN US TO DEVELOP

Next-generation Liquid-Cooling





ANNOUNCING
NEW WORKSTREAM LAUNCH

Tessera - Integrated Hardware Security Module (IHSM) Specification

Move cryptographic operations closer to workloads

Owner-Controlled Flow

HSM-backed security boundaries

Hardware-rooted trust

Building an Open IHSM Ecosystem: Standards, Software, and Collaboration

Publish Specification

With IHSM spec, we establish a standard to govern operations

Open Architecture

Define the integrated HSM architecture, requirements, protocol for interoperability.

Open Implementation

Develop open software, hardware, and firmware in a public GitHub repository

ASIC Path

Use a Caliptra™-like model, partner ecosystem of available chips, open reference implementation.

INDUSTRY OUTCOME: Standardized interoperability across integrated hardware security

Partnership on Tessera

ORACLE

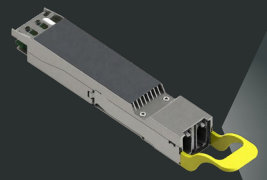
CALL TO ACTION: Join workgroup and support spec adoption

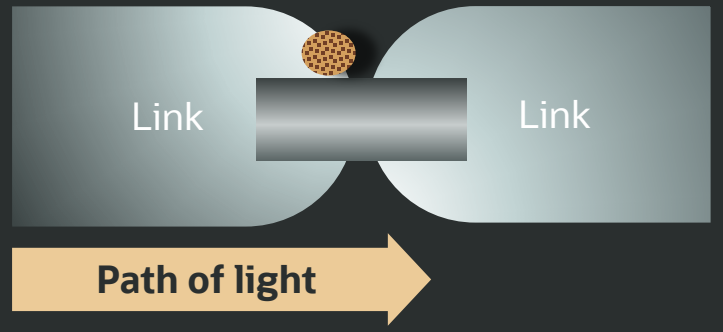
NETWORKING SPECIFICATIONS

Optics Reliability Specifications

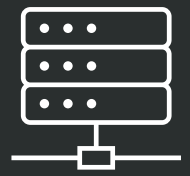
Enhanced Telemetry Optics Product Software Specification version 1.0 spec submitted to OCP

Network reliability is directly dependent on the network links





Dust on optics cause link failures



Millions of Links

Remediating link failures can take time

OCI drives open standards and collaboration for link reliability

Contribute

Bring the hardest lessons from operating hyperscale OCI fabrics into OCP

Standardize

Define vendor-neutral telemetry, link behavior, diagnosis, and mitigation

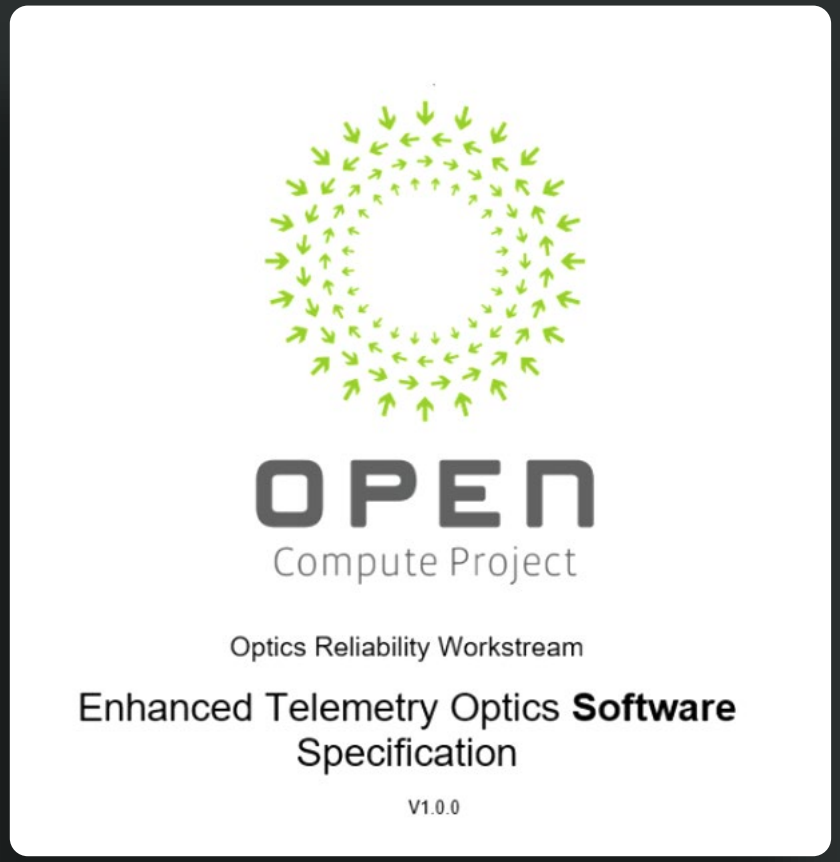
Catalyze

Give silicon, optics, systems, and operators a common foundation for faster innovation as LPO and CPO emerge.

Elevate

Raise interoperability, serviceability, and reliability across the industry.

INDUSTRY OUTCOME: A higher optics reliability bar for everyone



Link reliability is now a fabric-scale requirement

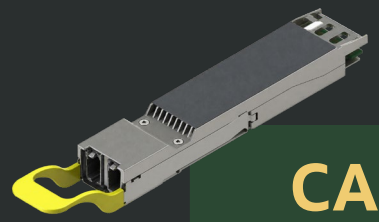
At OCI scale, isolated optical-link transients' cascade into fabric reconvergence, RDMA timeouts, and workload disruption.

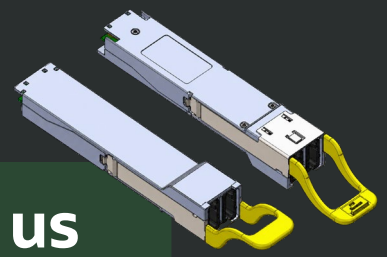
OCI proudly collaborates with our partners on the Optics Reliability initiative

Partnership implementing Optics Reliability spec. version 1.0

ORACLE CREDO MARVELL™

ARISTA BROADCOM eoptolink® COHERENT



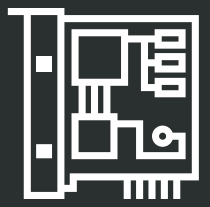


CALL TO ACTION: Connect, Collaborate, and Partner with us

ANNOUNCING

DPU Abstraction Layer

Architecture for Programmable
Networking Infrastructure Offload



DPU Abstraction Layer

OPEN BY DESIGN

A Reusable Platform Model

Preserves Innovation & Differentiation

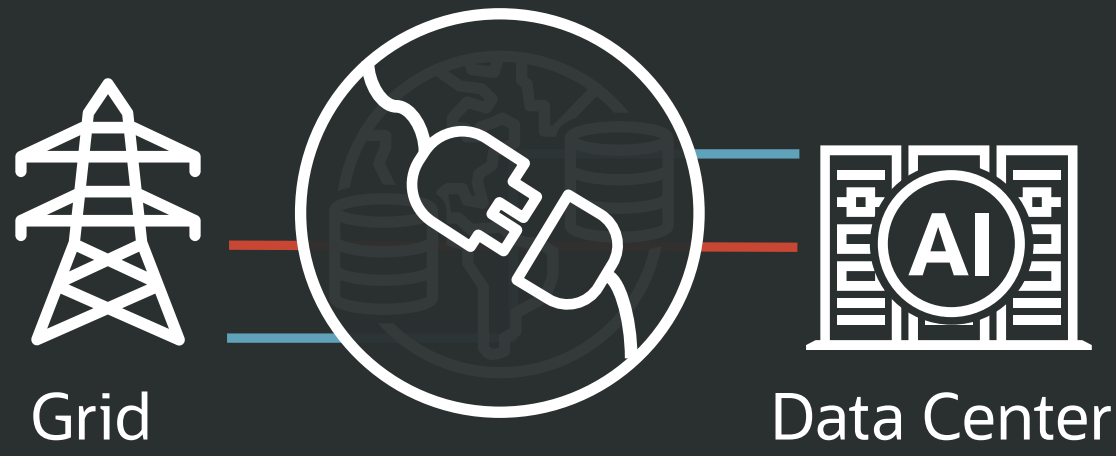
Open Architecture Alternative

Proven Production Model Ecosystem

CALL TO ACTION: Join us in defining next-gen interoperability

Energy and Grid Solutions Subgroup

- Uniform specs across the network of grid & energy industry partners, and DC providers
- Join us in the Energy and Grid Solutions for Grid-to-DC specs
- Additionally, OCI is working with the US Department of Energy & National Labs towards this common goal as well



Call to action

Standardize Responsible DC Citizenship

Prosperous Communities

Drive community-focused data center standards, partnering with OCP

Dynamic Data Center Architecture

Collaborate for an integrated framework for the industry

AI hardware specs for the DC of future

Accelerate interoperability with specs across - liquid cooled GPU, CPU, DIMMs, racks and network

Oracle's 2026 OCP Global Summit Speaker Schedule

Time	Talk	Speakers
Wed, Oct 14 | 8:55-9:10am	Lessons learned from GW-scale AI deployments	Mark Filer
Wed, Oct 14 | 3:55-4:10pm	Photonics for AI - Aligning the Stack from Components to Cloud	Stephen Manley
Thu, Oct 15 | 8:20-8:40am	Integrated HSM: Bringing Silicon-Level Key Security to the Server Tray	Mohan Kumar, Craig Barner, Vishal Soni
Thu, Oct 15 | 8:45-9:05am	Zero-Touch OCS Fabrics for AI Scale-Out Infrastructure at Hyperscale	Kannan Raj, Michael DeMerchant
Thu, Oct 15 | 9:10-9:30am	DCEM workstream update: Case studies of metric measurements from servers to data centers	Babak Falsafi; Murugasamy Nachimuthu; Sadas Shankar
Thu, Oct 15 | 9:45-10:00am	Sustainability and best practices for water efficiency	Rishab Vardhan Harikrishnan; Derek Kauneckis
Thu, Oct 15 | 9:45-10:10am	USB to Rule Them All: One Manageability Interface for AI Platforms	Mariusz Oriol; Bharat Pillili; Samer El-Haj-Mahmoud; Mohan Kumar; Supreeth Venkatesh; Kasper Wszolek
Thu, Oct 15 | 11:15-11:30am	Customer-Contextual Fault Management for Bare Metal GPU Infrastructure	Aimer Bhat; Jim Britton
Thu, Oct 15 | 12:30-12:40pm	Open Platform Firmware (OPF) - project update	Murugasamy Nachimuthu; Raj Kapoor
Thu, Oct 15 | 2:45-3:15pm	Scaling Memory Safe Security Across Firmware Stack	Giri Mudusuru; Srini Narayana; Tim Lewis; Albert Rennie; Bertrand Achard; Murugasamy Nachimuthu

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